UNIVERSITY OF ZAMBIA

CSC 6302 — Computer Vision

Practical 2 Report: Features that Matter

Student: Munyeme Sinyani (SINYANI)
Computer number: 2025024319
Photos: checkerboard; campus courtyard; agricultural show (pano crops)

Task 1 — Harris corners from scratch

I implemented Harris myself: Sobel gradients → I_x^2, I_y^2, I_xI_y Gaussian-smoothed → $R = \det(M) - k \cdot \mathrm{trace}(M)^2$ → relative threshold → NMS. No `cv2.cornerHarris`. Compared with Shi-Tomasi and FAST on a checkerboard and my campus courtyard photo. (FAST on the synthetic board needs a tiny blur — perfectly flat squares give it nothing to score.)

Image	Detector	Count	Time (ms)
checkerboard	harris	188	35.39
checkerboard	shi-tomasi	70	3.87
checkerboard	fast	190	0.21
natural	harris	1177	150.08
natural	shi-tomasi	500	33.28
natural	fast	12897	3.04

Comparison. Harris and Shi-Tomasi land on real square corners of the board; FAST is fastest. On the natural scene FAST fires densely on foliage/fence texture (~13k pts, ~3 ms); Harris is slower and more selective; Shi-Tomasi is capped at 500. Speed ranking: FAST ■ Shi-Tomasi > my Harris.

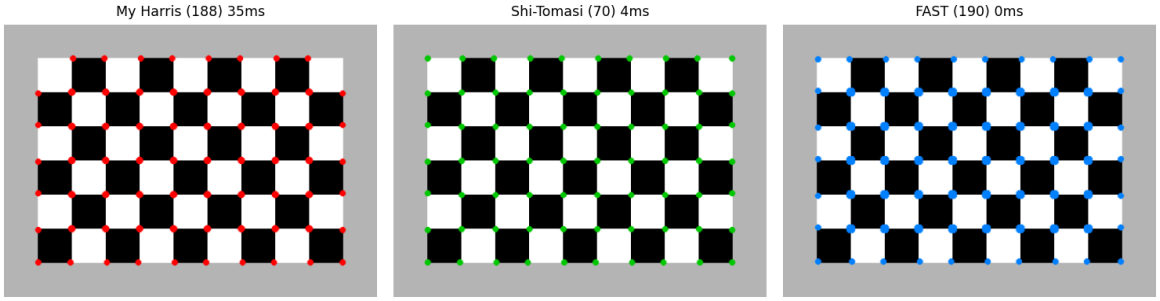


Figure 1. Checkerboard: Harris / Shi-Tomasi / FAST.

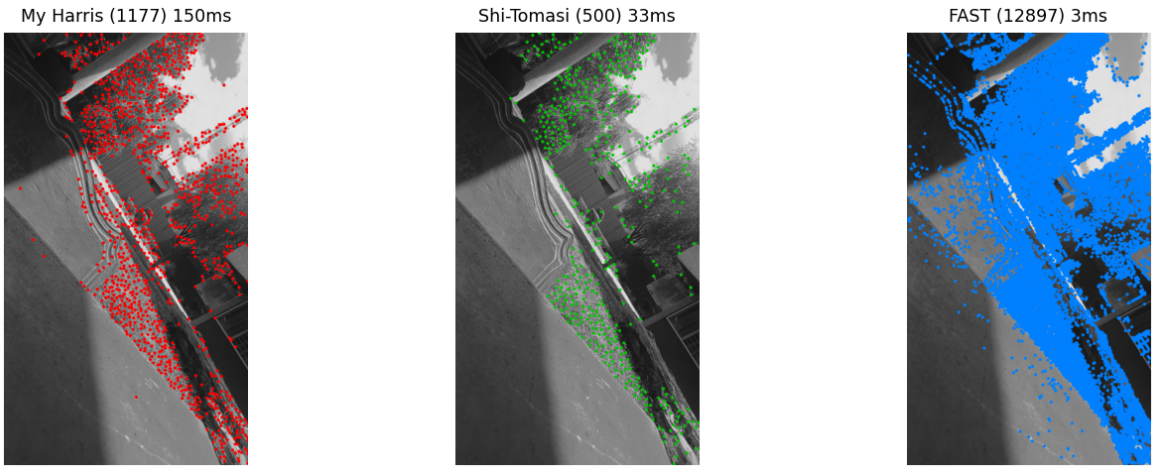


Figure 2. Campus courtyard: three detectors.

Task 2 — Descriptors & invariance

ORB and SIFT on my agricultural-show photo; repeatability after +30°, +90° and 0.5× scale (correct if within 3 px of the predicted location).

Detector	Transform	Repeatability %
ORB	rot30	30.92
ORB	rot90	43.25
ORB	scale05	24.25
SIFT	rot30	49.42
SIFT	rot90	62.58
SIFT	scale05	20.92

Ranking. SIFT wins both rotations (~63% vs ~43% at 90°) thanks to orientation assignment and gradient histograms. Scale 0.5× hurts both (empty borders on a fixed canvas); ORB is slightly ahead there in this particular test.



Figure 3. ORB vs SIFT keypoints.

Task 3 — Matching with the ratio test

SIFT + brute-force between overlapping show crops. Raw NN is messy; Lowe's ratio cleans it. Counts: ratio=0.6: matches=193; ratio=0.75: matches=218; ratio=0.9: matches=404. Lower ratio → fewer but safer matches; 0.9 keeps junk that can confuse RANSAC.



Figure 4. Raw NN vs ratio 0.75.

Task 4 — Panorama from a homography

Homography via RANSAC, `warpPerspective`, feather blend in the overlap (not `cv2.Stitcher`). `pair_left_mid_ransac_inliers=199 | blend=feather_average_in_overlap | third_image_ransac_inliers=436 |`.



Figure 5. Two-image panorama.



Figure 6. Three-crop wider panorama.

Task 5 — Robustness study

Degradation	Good matches	Inliers	Survived?
none	218	199	yes
blur	43	20	yes
dark60	65	51	yes
jpeg10	92	78	yes

Blur hurts most (gradients flatten). Darkening and JPEG also cut matches, but enough inliers remain for the panorama to survive — RANSAC can still estimate geometry from a thinner set of good matches.

Reproducibility & AI

`python prac2/prepare_images.py` then `python prac2/run_prac2.py`. Composer helped with stitching canvas ideas; Harris and interpretations are mine to explain.